Åbo Akademi University

Doctoral Programme in Biosciences and Drug Research

Åbo Akademi University

https://aaugs-dp-biosciences-and-drug-research.github.io/Home/

Scan for the website

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Maintained by Guillaume Jacquemet

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Learning goals

Online version

Learning goals for the Doctoral Programme in Biosciences and Drug Research

- Develop a deep understanding of the chosen field, including its core theories and concepts, their methodologies, as well as their social relevance.
- Design and execute innovative research projects using cutting-edge methods that lead to novel insights, development of theory and concepts and to practical applications.
- Build skills in problem-solving, project and time management, graphical design, scientific writing, communication, networking, teaching, supervision, leadership and other qualifications for demanding and multidisciplinary research- and expert positions in Finland and abroad.
- Prepare for diverse careers in academia, industry, the public sector, and beyond while fostering lifelong learning and leadership.

A doctoral degree unlocks many exciting career opportunities. Graduates actively shape innovations within academia and across public and private sectors, in Finland and globally. They drive breakthroughs in biotech and pharmaceutical companies, serve as influential experts in government ministries and research agencies, and lead transformative research at top-tier institutes. Additionally, they safeguard intellectual property at patent offices, steer higher education administration, and excel in scientific publishing, journalism, and entrepreneurial ventures.

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Lärandemål för doktorandprogrammet i biovetenskap och läkemedelsforskning

- Att utveckla en djup förståelse för det valda området, inklusive dess centrala teorier och koncept, deras metoder samt deras samhälleliga relevans.
- Att utforma och genomföra innovativa forskningsprojekt med hjälp av banbrytande metoder som leder till nya insikter, utveckling av teori och koncept samt till praktiska tillämpningar.
- Att bygga upp färdigheter i problemlösning, projekt- och tidshantering, grafisk design, vetenskapligt skrivande, kommunikation, nätverkande, undervisning, handledning, ledarskap och andra kvalifikationer för krävande och tvärvetenskapliga forsknings- och expertpositioner i Finland och utomlands.
- Att förbereda sig för olika karriärer inom den akademiska världen, industrin, den offentliga sektorn och utanför, samtidigt som livslångt lärande och ledarskap främjas.

En doktorsexamen möjliggör många spännande karriärmöjligheter. De utexaminerade forskarna formar aktivt innovationer inom den akademiska världen och inom den offentliga och privata sektorn, i Finland och globalt. De driver genombrott i bioteknik- och läkemedelsföretag, fungerar som inflytelserika experter i ministerier och forskningsinstitut och leder transformativ forskning vid toppuniversitet. Dessutom skyddar de immateriella rättigheter vid patentbyråer, styr administrationen av högre utbildning och utmärker sig inom vetenskaplig publicering, journalistik och entreprenörskap.

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Curriculum

Online version

The doctoral degree includes **30 ECTS** of studies. Most credits normally come from the major subject or a closely related field.

Major subject or related fields: 20–30 ECTS

Compulsory courses

- A course on research ethics
- Subject-specific compulsory courses

Courses that support the PhD project

Relevant options include advanced courses in the major subject or related fields and methodological training such as statistics, experimental design, or bioinformatics.

Conferences: maximum 5 ECTS

Five ECTS requires active participation, including the doctoral researcher's own contribution, in at least two international conferences.

General skills or minor subject: 0–10 ECTS

Examples include:

- University pedagogy
- Language courses, up to **5 ECTS**

Studying at other universities

Studies can also be completed at other universities. Through the **Doctoral+ (Tohtori+)** network, doctoral courses at other Finnish universities are free of charge; add the course to your study plan in Peppi.

A separate agreement also covers University of Turku courses outside the Doctoral+ network. See Doctoral studies at ÅAU for current information.

Licentiate degree

A licentiate degree consists of 30 ECTS of studies and a licentiate thesis. A doctoral degree completed after a licentiate degree only requires the doctoral dissertation. See the programme page for details.

Useful resource

- 📝 Skills and Competencies (DOCX)

Last updated: 28 September 2026

Online version

Yearly follow-up during your PhD studies

The yearly follow-up is a chance to review progress, update the study and research plans, and discuss supervision openly. The process combines the Individual Study Plan (ISP) with a Thesis Follow-Up Committee (TFC).

At the start of the PhD

Apply for the study right via Studieinfo.fi.

After admission, update the ISP within the first three months using the plan submitted with the doctoral-study application.

Every doctoral researcher has at least two supervisors. One is the responsible supervisor and must be a docent or have equivalent qualifications. See section 4.4 of the Regulation on Doctoral Studies.

Within the same first three months:

- recruit two TFC members together with the supervisor(s); and
- arrange the first TFC meeting.

The yearly cycle

The ISP must be updated once a year, normally at the end of the autumn semester unless the doctoral programme decides otherwise. The responsible supervisor also holds a follow-up and development meeting with the doctoral researcher at least once a year. See section 4.3 of the Regulation on Doctoral Studies.

The TFC meets every year until the thesis defence. The doctoral researcher organises the meetings.

The recommended committee has two members, internal or external to Åbo Akademi University, in addition to the supervisor(s). Its role is to provide scientific and practical advice and to help identify problems early. The study plan or research plan can be updated after the discussion.

If a problem with the research or supervision arises between meetings, the doctoral researcher can contact the TFC members or the ombudsperson. The researcher must have an opportunity to speak with the TFC in confidence.

Before each TFC meeting

At least one week before the meeting:

- update the ISP, including the research plan;
- send the updated ISP to the TFC and supervisors; and
- circulate the agenda and any other material.

During the meeting

Allow up to **90 minutes**.

1. **Researcher presentation and discussion — up to 60 minutes.** Cover the doctoral training plan and the main points needed to judge progress toward graduation:
   - completed and planned courses;
   - scientific progress;
   - publication plan;
   - funding plan; and
   - expected graduation date.
2. **Committee feedback.** Discuss the graduation plan, research plan, and presentation.
3. **Confidential discussion with the supervisor(s) — 10–15 minutes.** The doctoral researcher leaves the room so the committee and supervisor(s) can discuss any concerns.
4. **Confidential discussion with the doctoral researcher — final 10 minutes.** The supervisor(s) leave so the doctoral researcher can raise supervision-related questions or concerns directly with the committee.

After the meeting

Within one week, write short minutes, share them with the participants, make any final changes to the ISP, and submit the updated ISP to the graduate school.

Quick checklist

First three months

☐ Update the ISP
☐ Recruit two TFC members with the supervisor(s)
☐ Arrange the first TFC meeting

Before each yearly meeting

☐ Update the ISP and research plan
☐ Send them to the TFC and supervisor(s) at least one week in advance
☐ Circulate the agenda and supporting material

During each meeting

☐ Review graduation requirements and target date
☐ Review courses and scientific progress
☐ Review the publication plan
☐ Review the funding plan
☐ Hold the confidential discussion without the doctoral researcher
☐ Hold the confidential discussion without the supervisor(s)

Useful resource

- 📝 Skills and Competencies (DOCX)

Last updated: 28 September 2026

🎓 Graduation requirements

Online version

Duration and credits

A PhD thesis typically takes **3–4 years** to complete.

The degree requires **30 ECTS** of studies. See the Curriculum.

Article-based dissertations: university rules

Under the current Åbo Akademi University regulations, an article-based doctoral dissertation typically consists of **2–3 scientific articles**. A licentiate dissertation must include **at least one article**. Doctoral programmes may also provide programme-specific guidance.

The following can count as scientific articles:

- published articles;
- articles accepted for publication;
- unpublished manuscripts that have undergone peer review or been accepted for publication; and
- unpublished manuscripts submitted for peer review.

The articles form the main scientific contribution of the dissertation. The appropriate number depends on their scope, scientific significance and quality, the publication platform, and the doctoral researcher's individual contribution to co-authored work.

At least **one article must be accepted for publication**. In exceptional cases, the Head of Department may approve submission with fewer articles than normally recommended if the scope and quality are sufficient.

Preliminary examiners will pay particular attention to manuscripts that have not yet been accepted for publication.

Programme recommendations

The programme recommends judging a thesis by the quality and contribution of the work rather than by article count alone.

- A typical thesis contains **two high-quality papers**, with the doctoral researcher as **first author on at least one**.
- At least **50% of the included work** should be published or accepted.
- The supervisor and head of subject decide when the thesis is ready for preliminary examination.
- The head of department grants permission for the public defence on the basis of the preliminary examiners' statements.
- For every manuscript included in the thesis, the doctoral researcher's contribution should be described precisely. List concrete contributions such as data generation, analysis, writing, and relevant figure panels rather than using broad statements. Use the programme template signed by the supervisor.

Full regulations

- 📄 ÅAU Regulation on Doctoral Studies (PDF)

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Supervisor Portfolio for the Doctoral Programme in Biosciences and Drug Research

Online version

Biochemistry

KD

Käthe Dahlström



Peter Mattjus
Lipid Transfer Protein Research



Annika Meinander
Inflammatory Signalling Group



Tiina Salminen
Structural Immunology Lab

Cell Biology



Eleanor Coffey
Protein Kinase Function in the Brain

SG

Silvia Gramolelli
Viral Oncogenesis



Guillaume Jacquemet
Cell Migration Lab



Cecilia Sahlgren
Cellfate lab



Lea Sistonen
Sistonen Lab



Diana Toivola
Epithelial Biology Lab

Marine Biology

KA

Katri Aarnio

CB

Christoffer Boström
Boström's Lab



Mikael von Numers
Mikael von Numers



Christian Pansch-Hattich
Experimental Ecology - Stress Ecology & Ecophysiology



Riikka Puntila-Dodd
Marine ecosystem ecology



Sonja Salovius-Lauren
BiodiverSea-group



Conny Sjöqvist
Molecular Ecology Lab

MS

Martin Snickars
Husö



Anna Törnroos-Remes
Environmental and Marine Biology



Markus Öst
Eider Research Team

Pharmacy



Frederick Ahen
Global Pharmacopolitics



Kuldeep Bansal
Polymeric drug delivery

AF

Anne Filppula
Translational Drug Research



Zhang Hongbo
Functional Materials for Medicine



Jessica Rosenholm
BioNanoMaterials group



Outi Salo-Ahen
Computer-aided drug design (CADD) group



Tapani Viitala
Pharmaceutical Biophysics



Xiaoju Wang
Biomaterials and Drug Fabrication

Last updated: 10 November 2026

KD

Käthe Dahlström

Subject: Biochemistry
University: Åbo Akademi University
AboCRIS Profile: research.abo.fi/fi/persons/k%C3%A4the-dahlstr%C3%B6m

Online version

Last updated: 30 September 2026



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Peter Mattjus

Group Name: Lipid Transfer Protein Research
Subject: Biochemistry
University: Åbo Akademi University
Lab Website: users.abo.fi/pmattjus
AboCRIS Profile: research.abo.fi/sv/persons/peter-mattjus

Online version

Areas of Expertise

- The role of glycolipid transfer proteins in cells
- Glycosphingolipid biosynthesis
- Intracellular glycosphingolipid transport events

Research Projects

- Glycolipid transfer protein, GLTP interaction with VAP-proteins in the ER exit sites
- The role of GLTP in demyelination
- Intracellular localization of GLTP
- Inhibitors for glycolipid binding proteins

Special Methodologies & Techniques

- Lipid biochemistry
- Quantitative and qualitative glycosphingo- and phospholipid analysis by high-performance TLC
- Cell and molecular biology
- Fluorescence spectroscopy, steady-state and lifetime
- Radioisotope labeling and lipid metabolism techniques

Major Funding & International Networks

- Medicinska understödsföreningen Liv och hälsa
- Borgs stiftelse
- COST action SPHINX member

Selected Publications

- Glycolipid transfer protein knockout disrupts vesicle trafficking to the plasma membrane. Journal of Biological Chemistry. 2023. DOI: 10.1016/j.jbc.2023.104607
- Who moves the sphinx? An overview of intracellular sphingolipid transport. Biochimica et Biophysica Acta (BBA) - Molecular and Cell Biology of Lipids. 2021. DOI: 10.1016/j.bbalip.2021.159021
- Glucosylceramide acyl chain length is sensed by the glycolipid transfer protein. PLOS ONE. 2018. DOI: 10.1371/journal.pone.0209230
- Alternation in the Glycolipid Transfer Protein Expression Causes Changes in the Cellular Lipidome. PLoS ONE. 2014. DOI: 10.1371/journal.pone.0097263
- Vesicular and non-vesicular transport feed distinct glycosylation pathways in the Golgi. Nature. 2013. DOI: 10.1038/nature12423

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Annika Meinander

Online version

Group Name: Inflammatory Signalling Group
Subject: Biochemistry
University: Åbo Akademi University
Lab Website: abo.fi/en/inflammatory-signalling-lab
AboCRIS Profile: research.abo.fi/sv/persons/annika-meinander

Areas of Expertise

- Post-translational modifications, ubiquitination in particular
- NF-κB signalling pathways
- Apoptotic and non-apoptotic roles of caspases
- Drosophila melanogaster as a model system

Research Projects

- Ubiquitination-mediated regulation of inflammation during cell stress and infection
- Targeting ubiquitin signalling in chronic inflammation and inflammation-induced cancer
- Ubiquitin-mediated control of starvation-induced intestinal inflammatory responses
- Caspase-mediated regulation of autophagy in inflammatory signalling
- Modulating caspase activation and immune signalling in intestinal inflammation

Special Methodologies & Techniques

- Analysis of ubiquitin modifications in vitro and in vivo
- Drosophila melanogaster as a model system to study cell signalling and immune responses
- Imaging caspase activation in vivo

Major Funding & International Networks

- InFLAMES Flagship
- Sigrid Jusélius Foundation
- Swedish Cultural Foundation
- International Society of Protein Termini

Selected Publications

- M1-linked ubiquitination by LUBAC regulates AMPK signalling and the response to energetic stress. Cell Death & Differentiation. 2026. DOI: 10.1038/s41418-026-01675-z
- Ubiquitin signalling in *Drosophila* innate immune responses. The FEBS Journal. 2024. DOI: 10.1111/febs.17028
- Drosophila caspases as guardians of host-microbe interactions. Cell Death & Differentiation. 2023. DOI: 10.1038/s41418-022-01038-4
- M1-linked ubiquitination facilitates NF-κB activation and survival during sterile inflammation. The FEBS Journal. 2022. DOI: 10.1111/febs.16425
- Drice restrains Diap2-mediated inflammatory signalling and intestinal inflammation. Cell Death & Differentiation. 2022. DOI: 10.1038/s41418-021-00832-w

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Tiina Salminen

Group Name: Structural Immunology Lab
Subject: Biochemistry
University: Åbo Akademi University
Lab Website: abo.fi/en/structural-immunology-lab
AboCRIS Profile: research.abo.fi/en/persons/tiina-a-salminen

Online version

Areas of Expertise

- Protein biochemistry, bioinformatics, and Structural biology
- Protein-ligand interactions
- Molecular interaction mechanisms in inflammation, infection, and cancer
- Disease-causing genetic variation
- Structure-based drug design
- Targeted diagnostics

Research Projects

- Selective treatments against Borrelia infections
- Immunosuppressive SEFIR protein asa target against Klebsiella infections
- Expanding toolbox for Clever-1-targeted oncotherapy
- Intercepting Siglec-9 interactions in cancer and inflammation
- Molecular mechanism of collagen-binding integrin activation

Special Methodologies & Techniques

- Modern facilities for protein purification and sample quality control
- State-of-the-art infrastructure for protein crystallization
- Automated imaging of protein crystallization trials
- In-situ diffraction testing
- Infrastructure for Cryo-electron microscopy sample preparation
- Up-to-date infrastructure for computational biology and Scientific IT Support

Major Funding & International Networks

- Research Council Finland
- InFLAMES Flagship
- Medicinska Understödsföreningen Liv och Hälsa rf
- Tor, Joe och Pentti Borg Foundation
- 3D-BioInfo Community of ELIXIR
- International Network of Protein Engineering Centres (INPEC)

Selected Publications

- Secreted Clever-1 modulates T cell responses and impacts cancer immunotherapy efficacy. Theranostics. 2025. DOI: 10.7150/thno.110544
- Apprehensions and emerging solutions in ML-based protein structure prediction. Current Opinion in Structural Biology. 2024. DOI: 10.1016/j.sbi.2024.102819
- Vascular adhesion protein-1-targeted PET imaging in autoimmune myocarditis. Journal of Nuclear Cardiology. 2023. DOI: 10.1007/s12350-023-03371-8
- Klebsiella pneumoniae type VI secretion system-mediated microbial competition is PhoPQ controlled and reactive oxygen species dependent. PLOS Pathogens. 2020. DOI: 10.1371/journal.ppat.1007969
- Structural and Biomolecular Analyses of Borrelia burgdorferi BmpD Reveal a Substrate-Binding Protein of an ABC-Type Nucleoside Transporter Family. Infection and Immunity. 2020. DOI: 10.1128/IAI.00962-19

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Eleanor Coffey

Online version

Group Name: Protein Kinase Function in the Brain
Subject: Cell Biology
University: Åbo Akademi University
Lab Website: bioscience.fi/research/kinase-function-in-brain
AboCRIS Profile: research.abo.fi/en/persons/eleanor-coffey

Areas of Expertise

- Our lab investigates the molecular mechanisms underlying brain disorders, with specialized expertise in the signaling pathways regulated by the protein kinases JNK and LRRK2. We have uncovered novel insights into how these kinases influence anxiety-like behaviors and neuronal death in Parkinson's disease. By integrating animal and cellular models with clinical proteomics, we aim to deepen our understanding of the biology behind these conditions.

Research Projects

- We focus on Parkinson's disease, using deep proteomic profiling of blood cells to detect pre-symptomatic changes. This includes developing new informatics to decode the "dark proteome" of cohorts like PPMI to identify early biomarkers. We also study depression cohorts to find markers predicting treatment response. Together, this work offers insight into immune cell roles in brain disorders.

Special Methodologies & Techniques

- Our lab combines fluorescence imaging and proteomics methods with neurobiology, biochemistry, and data science to reach our goals.

Major Funding & International Networks

- These projects are funded by the Michael J Fox Foundation, the Research Council of Finland, EU ERANET, Immudocs, and Inflames. We participate in international networks in the field of neuroscience, including informatics for neuroscience.

Selected Publications

- PhosPiR: an automated phosphoproteomic pipeline in R. Briefings in Bioinformatics. 2022. DOI: 10.1093/bib/bbab510
- Protein synthesis is suppressed in sporadic and familial Parkinson's disease by LRRK2. The FASEB Journal. 2020. DOI: 10.1096/fj.202001046R
- JNK1 controls adult hippocampal neurogenesis and imposes cell-autonomous control of anxiety behaviour from the neurogenic niche. Molecular Psychiatry. 2018. DOI: 10.1038/mp.2016.203
- Nuclear and cytosolic JNK signalling in neurons. Nature Reviews Neuroscience. 2014. DOI: 10.1038/nrn3729
- Phosphorylation of SCG10/stathmin-2 determines multipolar stage exit and neuronal migration rate. Nature Neuroscience. 2011. DOI: 10.1038/nn.2755

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SG

Silvia Gramolelli

Online version

Group Name: Viral Oncogenesis
Subject: Cell Biology
University: Åbo Akademi University
Lab Website: viraloncor.wordpress.com
AboCRIS Profile: research.abo.fi/sv/persons/silvia-gramolelli/publications

Areas of Expertise

- Gene expression
- Chromatin remodelling
- Viral reactivation from latency
- Cellular transformation

Research Projects

- Role of somatic mutations in Epstein-Barr virus-induced oncogenesis and drug resistance
- Role of stress-related transcription factors in oncogenic herpesvirus persistence and gene expression
- New diagnostic tools based on DNA nanotechnology to detect viral genomic sequences

Special Methodologies & Techniques

- Patient-derived organoids
- Lentiviral production and transduction
- Chromatin IP
- Proteomics

Major Funding & International Networks

- Research Council of Finland
- Sigrid Juselius
- Finnish Cultural Foundation
- Mary & George Ehrnrooth Foundation
- Suomen Tiedeseura
- InFLAMES

Selected Publications

- Heat shock factor 2 regulates oncogenic gamma-herpesvirus gene expression by remodeling the chromatin at the ORF50 and BZLF1 promoter. PLOS Pathogens. 2025. DOI: 10.1371/journal.ppat.1013108
- DLL4/Notch3/WNT5B axis mediates bidirectional prometastatic crosstalk between melanoma and lymphatic endothelial cells. JCI Insight. 2024. DOI: 10.1172/jci.insight.171821
- Oncogenic Herpesvirus Engages Endothelial Transcription Factors SOX18 and PROX1 to Increase Viral Genome Copies and Virus Production. Cancer Research. 2020. DOI: 10.1158/0008-5472.CAN-19-3103
- Kaposi's Sarcoma-Associated Herpesvirus Lytic Replication Is Independent of Anaphase-Promoting Complex Activity. Journal of Virology. 2020. DOI: 10.1128/JVI.02079-19
- High tissue MMP14 expression predicts worse survival in gastric cancer, particularly with a low PROX1. Cancer Medicine. 2019. DOI: 10.1002/cam4.2576

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Guillaume Jacquemet

Online version

Group Name: Cell Migration Lab
Subject: Cell Biology
University: Åbo Akademi University
Lab Website: cellmig.org
AboCRIS Profile: research.abo.fi/en/persons/guillaume-jacquemet
Web Pages: ORCID · Google Scholar

Areas of Expertise

- Cancer cell migration, invasion, and metastasis
- Filopodia, cell adhesion, and cytoskeletal dynamics
- Endothelial barriers and cancer-endothelium interactions
- Advanced microscopy and quantitative live-cell imaging
- Bioimage analysis, cell tracking, deep learning, and computer vision

Research Projects

- Mechanisms of pancreatic cancer arrest and extravasation at vascular barriers
- Filopodia in cancer invasion and cell-cell communication
- Endothelial barrier regulation and immune-endothelial interactions
- FlowVision: label-free imaging and quantitative analysis of cell arrest under flow
- Accessible deep learning and reproducible bioimage analysis for microscopy
- Tracking, registration, and quantitative microscopy tools

Special Methodologies & Techniques

- Live-cell, spinning-disk, and super-resolution microscopy
- Microfluidic flow, perfusion, and endothelial barrier assays
- Quantitative image analysis, cell tracking, segmentation, and computer vision
- Deep learning for microscopy and reproducible bioimage-analysis workflows
- Cell and molecular biology and protein-interaction analysis
- Zebrafish and mouse models of cancer progression and metastasis

Major Funding & International Networks

- Research Council of Finland - Project Grant and IMMENs Centre of Excellence
- Finnish Cancer Institute - K. Albin Johansson Research Professorship
- Cancer Foundation Finland
- Wellcome Trust / Chan Zuckerberg Initiative - Essential Open Source Software for Science (EOSS)
- Sigrid Juselius Foundation

Selected Publications

- CCT8 associates with the MYO10 motor domain and regulates filopodia and breast cancer cell invasion. Journal of Cell Science. 2026. DOI: 10.1242/jcs.265278
- Fast label-free live imaging with FlowVision reveals key principles of cancer cell arrest on endothelial monolayers. The EMBO Journal. 2026. DOI: 10.1038/s44318-025-00678-9
- TLNRD1 is a CCM complex component and regulates endothelial barrier integrity. Journal of Cell Biology. 2024. DOI: 10.1083/jcb.202310030
- DL4MicEverywhere: deep learning for microscopy made flexible, shareable and reproducible. Nature Methods. 2024. DOI: 10.1038/s41592-024-02295-6
- CellTracksColab is a platform that enables compilation, analysis, and exploration of cell tracking data. PLOS Biology. 2024. DOI: 10.1371/journal.pbio.3002740

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Cecilia Sahlgren

Group Name: Cellfate lab
Subject: Cell Biology
University: Åbo Akademi University
Lab Website: cellfatelab.squarespace.com
AboCRIS Profile: research.abo.fi/en/persons/cecilia-sahlgren

Online version

Areas of Expertise

- Cell biology of soft tissue development, regeneration and disease (muscle, cardiovascular)
- Cell Signaling and Mechanobiology
- Microtissue engineering

Research Projects

- Immune-endothelial interphases (IMMENs)
- Hematopoietic stem cell engineering (Making blood)
- Mechanobiology of hematopoietic stem cell niches and the endothelium
- Deciphering mechanisms of soft tissue development and regeneration
- Nanoparticle interactions with biological barriers
- Deciphering mechanisms of tumor-endothelial crosstalk

Special Methodologies & Techniques

- Microtissue engineering and biomaterials
- Organ on Chips, flow and perfusion chambers, micropatterning, functionalized materials
- Cell and molecular biology
- Optogenetic tools and synthetic biology tools
- Zebrafish to study soft tissue development (muscle, cardiovascular)
- Computational simulations of mechanochemical regulation of tissue architecture

Major Funding & International Networks

- Sigrid Juselius Foundation
- European Research Council and European Commission (H2024, Marie Curie Training Network)
- Research Council of Finland
- Inflames
- Turku Bioscience
- Materials-driven regeneration (https://mdrresearch.nl)
- Institute for complex molecular systems (https://www.tue.nl/en/research/institutes/institute-for-complex-molecular-systems)

Selected Publications

- An integrated 2D framework for quantifying cellular mechanics reveals the impact of juxtacrine Notch signalling on directed collective migration of endothelial cells. BMC Biology. 2025. DOI: 10.1186/s12915-025-02396-4
- Receptor stoichiometry predicts artery-typical vulnerability to altered Notch signaling during smooth muscle differentiation. bioRxiv. 2025. DOI: 10.1101/2025.05.05.652186
- Mechanosensitive interactions between Jag1 and Myo1c control Jag1 trafficking in endothelial cells. iScience. 2025. DOI: 10.1016/j.isci.2025.113879
- 2.5D Model for *Ex Vivo* Mechanical Characterization of Sprouting Angiogenesis in Living Tissue. Journal of Visualized Experiments. 2025. DOI: 10.3791/67641
- Engineered patterns of Notch ligands Jag1 and Dll4 elicit differential spatial control of endothelial sprouting. iScience. 2022. DOI: 10.1016/j.isci.2022.104306

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Lea Sistonen

Online version

Group Name: Sistonen Lab
Subject: Cell Biology
University: Åbo Akademi University
Lab Website: sistonenlab.com
AboCRIS Profile: research.abo.fi/sv/persons/lea-sistonen

Areas of Expertise

- Cell Differentiation
- Cell Stress
- Post-translational Modifications
- Proteostasis
- Transcriptional Gene Regulation

Research Projects

- HSF-NET: Functional networks regulated by heat shock factors in cell plasticity, Research Council of Finland 2023-27
- Proteostasis driven by HSFs in cell plasticity programs and cancer", Sigrid Jusélius Foundation 2025-28
- Re-programming of gene and enhancer expression in cell differentiation and stress, Liv och Hälsa 2023-26
- Pharmacologic strategy for targeting the main oncoproteins and defense systems of HPV-driven cancers, Cancer Foundation Finland 2024-26, together with John E. Eriksson

Special Methodologies & Techniques

- ChIP-seq
- DNA-protein interaction in vivo and in vitro
- Proteomics
- PRO-seq and RNA-seq, including data analysis

Major Funding & International Networks

- HSF-NET: Functional networks regulated by heat shock factors in cell plasticity, Research Council of Finland 2023-27
- Proteostasis driven by HSFs in cell plasticity programs and cancer", Sigrid Jusélius Foundation 2025-28
- Re-programming of gene and enhancer expression in cell differentiation and stress, Liv och Hälsa 2023-26
- Pharmacologic strategy for targeting the main oncoproteins and defense systems of HPV-driven cancers, Cancer Foundation Finland 2024-26, together with John E. Eriksson

Selected Publications

- Nuclear talin-1 provides a bridge between cell adhesion and gene expression. iScience. 2025. DOI: 10.1016/j.isci.2025.111745
- CBP-HSF2 structural and functional interplay in Rubinstein-Taybi neurodevelopmental disorder. Nature Communications. 2022. DOI: 10.1038/s41467-022-34476-2
- HSFs drive transcription of distinct genes and enhancers during oxidative stress and heat shock. Nucleic Acids Research. 2022. DOI: 10.1093/nar/gkac493
- Stress-induced transcriptional memory accelerates promoter-proximal pause release and decelerates termination over mitotic divisions. Molecular Cell. 2021. DOI: 10.1016/j.molcel.2021.03.007
- Heat Shock Factor 2 Protects against Proteotoxicity by Maintaining Cell-Cell Adhesion. Cell Reports. 2020. DOI: 10.1016/j.celrep.2019.12.037

Last updated: 30 September 2026



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Diana Toivola

Group Name: Epithelial Biology Lab
Subject: Cell Biology
University: Åbo Akademi University
Lab Website: toivola-lab.org
AboCRIS Profile: research.abo.fi/sv/persons/diana-toivola

Online version

Areas of Expertise

- Epithelial cell biology of gastrointestinal organs and related glands including the endocrine pancreas
- Cytoskeletal intermediate filaments protein regulation and function and molecular roles in health and disease, especially simple epithelial keratins and nuclear lamins
- Auto-immune diseases, Inflammatory bowel diseases, type I diabetes, and colorectal cancer
- Mouse disease modelling
- Non-invasive diagnostics development of intestinal diseases

Research Projects

- Dysregulation of keratins in intestinal inflammation with biomarker potential
- Noninvasive in vitro assays to detect and monitor inflammatory bowel diseases
- A colon-specific mouse model for colorectal cancer
- Keratins as tumor suppressors and guardians of intestinal health
- Beta-cell cytoskeletal keratin 18 and its putative type 1 diabetes-associated variant in beta-cell function

Special Methodologies & Techniques

- Mouse disease modelling, especially models of colitis, colorectal cancer and diabetes
- Intestinal crypt isolation and organoid culture (mini-gut)
- Isolation of islets of Langerhans and 3D culture
- Beta-cell and colon cancer cell lines
- Microscopy, digital histopathology and fluorescence-based imaging methods
- Cell, molecular, biochemical methods
- Immunoassay and non-invasive imaging methods development

Major Funding & International Networks

- Current funding: InFLAMES flagship, ImmuDoc, Medical Support foundation Liv och Hälsa, Diabetes Research Foundation Finland, Wilhelm and Else Stockman foundation
- Networks: Intermediate filament community, American Society for Cell Biology, Gut Research Community in Finland, Gastroenterology organizations in Finland, and USA

Selected Publications

- Genetically Induced Mouse Model for Colon-specific Epithelial Cell Tumorigenesis Driven by Loss of K8 and Apc. Cellular and Molecular Gastroenterology and Hepatology. 2026. DOI: 10.1016/j.jcmgh.2025.101716
- Defining a timeline of colon pathologies after keratin 8 loss: rapid crypt elongation and diarrhea are followed by epithelial erosion and cell exfoliation. American Journal of Physiology-Gastrointestinal and Liver Physiology. 2024. DOI: 10.1152/ajpgi.00140.2023
- β-Cell keratin 8 maintains islet mechanical integrity, mitochondrial ultrastructure, and β-cell glucose transporter 2 plasma membrane targeting. American Journal of Physiology-Cell Physiology. 2024. DOI: 10.1152/ajpcell.00123.2024
- Colonocyte keratins stabilize mitochondria and contribute to mitochondrial energy metabolism. American Journal of Physiology-Gastrointestinal and Liver Physiology. 2024. DOI: 10.1152/ajpgi.00220.2023
- Colonocyte keratin 7 is expressed de novo in inflammatory bowel diseases and associated with pathological changes and drug-resistance. Scientific Reports. 2022. DOI: 10.1038/s41598-022-26603-2

Last updated: 30 September 2026

KA

Katri Aarnio

Subject: Marine Biology
University: Åbo Akademi University
AboCRIS Profile: research.abo.fi/sv/persons/katri-aarnio

Online version

Last updated: 30 September 2026

CB

Christoffer Boström

Group Name: Boström's Lab
Subject: Marine Biology
University: Åbo Akademi University
AboCRIS Profile: research.abo.fi/sv/persons/christoffer-bostr%C3%B6m/projects

Online version

Areas of Expertise

- Marine ecology

Research Projects

- Digital Waters

Special Methodologies & Techniques

- Marine field sampling, mesocosm experiments, field experiments

Major Funding & International Networks

- Doctoral Pilot
- Svenska Kulturfonden
- networks Zostera Experimental Network
- Nordic collaboration

Selected Publications

- Marine biodiversity loss in Finnish coastal waters: Evidence and implications for management. Ambio. 2025. DOI: 10.1007/s13280-025-02185-x
- Shallow Coastal Bays as Sediment Carbon and Nutrient Reservoirs in the Baltic Sea. Estuaries and Coasts. 2025. DOI: 10.1007/s12237-025-01541-0
- Assessing the success of marine ecosystem restoration using meta-analysis. Nature Communications. 2025. DOI: 10.1038/s41467-025-57254-2
- The methylome of clonal seagrass shoots shows age-associated variation and differentiation of roots from other tissues. Biochimica et Biophysica Acta (BBA) - General Subjects. 2025. DOI: 10.1016/j.bbagen.2024.130748
- Climate Effects on Belowground Tea Litter Decomposition Depend on Ecosystem and Organic Matter Types in Global Wetlands. Environmental Science & Technology. 2024. DOI: 10.1021/acs.est.4c02116

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Mikael von Numers

Online version

Group Name: Mikael von Numers
Subject: Marine Biology
University: Åbo Akademi University
AboCRIS Profile: research.abo.fi/sv/persons/mikael-von-numers

Areas of Expertise

- Island biogeography
- Long-term changes
- Distribution patterns among plants and birds
- Species-environment interactions

Research Projects

- My main research interests include long-term changes (among both plants and birds) and biogeographical questions concerning species distribution patterns in relation to the environment.

Special Methodologies & Techniques

- Fieldwork, GIS

Major Funding & International Networks

- mainly domestic sources such as SKF

Selected Publications

- Predation risk and landscape properties shape reproductive output of an endangered sea duck from two subpopulations with contrasting predation risk. Journal of Ornithology. 2023. DOI: 10.1007/s10336-022-02036-6
- Island properties dominate species traits in determining plant colonizations in an archipelago system. Ecography. 2020. DOI: 10.1111/ecog.05013
- Population changes in the declining Turnstone (Arenaria interpres) and other waders in the northern Baltic Sea based on past and current breeding observations. Ornis Fennica. 2020. DOI: 10.51812/of.133973
- Refining predictions of metacommunity dynamics by modeling species non-independence. Ecology. 2020. DOI: 10.1002/ecy.3067
- Distribution Patterns and Long-Term Changes in Vascular Plants of Non-Littoral Areas in the SW Archipelago of Finland. Part VII. Poaceae and Synthesis. Annales Botanici Fennici. 2019. DOI: 10.5735/085.056.0418

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Christian Pansch-Hattich

Group Name: Experimental Ecology - Stress Ecology & Ecophysiology
Subject: Marine Biology
University: Åbo Akademi University
Lab Website: pansch-research.com
AboCRIS Profile: research.abo.fi/en/persons/christian-pansch-hattich

Online version

Areas of Expertise

- Climate change impacts on marine systems; environmental variability & extreme climatic events, heatwaves, climate change refugia, thermal microclimates
- Area: Ecophysiology, thermal ecology, acclimation & adaptation, species interactions e.g., facilitation, mesocosm food webs, invasion ecology, macrophyte-grazer interactions
- Systems: Bivalves, macrophytes (seagrass, Fucus), associated epi- and infauna, crustaceans (Gammarus, Idotea, mudcrabs), phytoplankton

Research Projects

- NEMESIS (Networked Environmental Monitoring and Evaluation System for Intelligent Surveillance) research infrastructure (RI) Thematic research infrastructures: Local research infrastructures to strengthen regional R&D
- ÅA Stiftelse: SOS - Centre for Sustainable Ocean Science
- Horizon: SEA-Quester - Blue Carbon production, export, & sequestration in emerging polar ecosystems
- Res. Counc. Norway: NORSE - Biodiversity in Northern European Seagrass meadows – drivers, responses, & resilience
- Collaborator within Susanne Kortsch's Academy fellowship that was funded by the Research Council of Finland for the project "Heatwave-driven rewiring of coastal food webs" at the University of Helsinki
- PhD L. Kuismanen: Drivers of underwater biodiversity patterns and targeting mitigation actions in a morphometrically complex archipelago – Integrating extensive high-resolution data
- PhD A. Skrifvars: Resilience of Arctic and Subarctic Phytoplankton to Rising Sea Surface Temperature
- PhD S. Rühmkorff: Resilience of Seagrass Ecosystems through Habitat Heterogeneity and Genetic Diversity
- PhD L. Kraufvelin: Impacts of marine heatwaves on the functioning of temperate coastal ecosystems under the influence of non-indigenous species

Special Methodologies & Techniques

- Combine ecophysiology, acclimation, and adaptation experimental studies with simplified community assessments in mostly experimental approaches
- Using high-throughput incubation units and mesocosms
- Simulating multiple and fluctuating climate change drivers
- High-resolution coastal environmental monitoring

Major Funding & International Networks

- Funders: EU, Research Council Fi, Svenska Kulturfonden, Stiftelse Åbo Akademi, City of Turku
- Collaborations: Tvärminne Zoological Station at Helsinki University, Turku, University of Applied Sciences – TUAS, Tjärnö Marine Laboratory at Gothenburg University, GEOMAR – Helmholtz Centre for Ocean Research Kiel, Alfred Wegener Institute for polar and marine research (AWI) – Wadden Sea Station Sylt, Leibniz Institute for Baltic Sea Research Warnemünde – IOW

Selected Publications

- Small-scale thermal habitat variability may not determine seagrass resilience to climate change. Limnology and Oceanography. 2025. DOI: 10.1002/lno.70049
- The interplay of co-occurring ecosystem engineers shapes the structure of benthic communities – a mesocosm experiment. Frontiers in Marine Science. 2024. DOI: 10.3389/fmars.2024.1304442
- Marine heatwaves and upwelling shape stress responses in a keystone predator. Proceedings of the Royal Society B: Biological Sciences. 2023. DOI: 10.1098/rspb.2022.2262
- Environmental variability in aquatic ecosystems: Avenues for future multifactorial experiments. Limnology and Oceanography Letters. 2023. DOI: 10.1002/lol2.10286
- Editorial: Influence of environmental variability on climate change impacts in marine ecosystems. Frontiers in Marine Science. 2022. DOI: 10.3389/fmars.2022.994756

Last updated: 10 November 2026



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Riikka Puntila-Dodd

Group Name: Marine ecosystem ecology
Subject: Marine Biology
University: Åbo Akademi University
AboCRIS Profile: research.abo.fi/en/persons/riikka-puntila-dodd

Online version

Areas of Expertise

- Ecosystem modelling
- Ecopath with Ecosim
- Species distribution modelling
- Future scenarios
- Baltic Sea
- Cumulative pressures

Research Projects

- GES4SEAS
- MIMOSA
- Identifying changes in coastal ecosystems – implications to
- Recovery trajectories (RCoF)
- SEADITO

Special Methodologies & Techniques

- Ecopath with Ecosim
- Bayesian networks
- Species distribution modelling
- Integrated trend analysis
- Multivariate analyses

Major Funding & International Networks

- RCoF
- Horizon Europe ICES WGIAB
- Marine modelling network
- Ecopath Consortium
- Beyond shifting baselines-consortium

Selected Publications

- Novelty, variability, and resilience: Exploring adaptive cycles in a marine ecosystem under pressure. Ambio. 2025. DOI: 10.1007/s13280-025-02181-1
- Food web robustness depends on the network type and threshold for extinction. Oikos. 2025. DOI: 10.1111/oik.11139
- Modelling framework to evaluate societal effects of ecosystem management. Science of The Total Environment. 2023. DOI: 10.1016/j.scitotenv.2023.165508
- Integrating diverse model results into decision support for good environmental status and blue growth. Science of The Total Environment. 2022. DOI: 10.1016/j.scitotenv.2021.150450
- Food-Web Modeling in the Maritime Spatial Planning Challenge Simulation Platform: Results from the Baltic Sea Region. Simulation Gaming Through Times and Disciplines. 2021. DOI: 10.1007/978-3-030-72132-9_25

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Sonja Salovius-Lauren

Online version

Group Name: BiodiverSea-group
Subject: Marine Biology
University: Åbo Akademi University
Lab Website: abo.fi/en/emb-research
AboCRIS Profile: research.abo.fi/sv/persons/sonja-salovius-lauren

Areas of Expertise

- Marine biodiversity and marine inventories
- Spatial ecology and conservation planning
- Marine restoration
- Sediment-biological interactions
- Genetic (molecular markers)
- Macroalgal-invertebrate interactions
- Marine indicator development

Research Projects

- Biodiversea-project, funded by EU LIFE IP
- Center for Sustainable Ocean Sciences (SOS), Center of Excellence, funded through ÅAU Endowment
- RIKI- project ("Nutrient turnover in the Archipelago Sea") in cooperation with the ELY Center, funded by the Ministry of the Environment
- The Finnish Inventory Programme for Underwater Marine Diversity, VELMU

Special Methodologies & Techniques

- Marine mapping methods, including UW video camera and photography
- Aquarium experiments
- Sediment analysis
- Molecular markers of selected species (charophytes, blue mussels)
- Spatial modeling

Major Funding & International Networks

- The present funding comes through EU Life, ÅAU endowment, and the Ministry of Environment/ELY-center. Recent funding: Baltic Sea Conservation Foundation, Government of Åland, European Maritime and Fisheries Fund, and the Ministry of Environment

Selected Publications

- Sediment Characteristics in Stonewort *Chara tomentosa* Assemblages. Ecology and Evolution. 2026. DOI: 10.1002/ece3.72772
- Water depth affects the biodiversity and functional trait composition of Fucus vesiculosus-associated macroinvertebrate communities in the Baltic Sea. Marine Ecology Progress Series. 2026. DOI: 10.3354/meps15029
- Untangling the literature on unattached *Fucus* (Phaeophyceae): an overview of current knowledge. European Journal of Phycology. 2025. DOI: 10.1080/09670262.2025.2532468
- Marine biodiversity loss in Finnish coastal waters: Evidence and implications for management. Ambio. 2025. DOI: 10.1007/s13280-025-02185-x
- Finnish inventory data of underwater marine biodiversity. Scientific Data. 2024. DOI: 10.1038/s41597-024-04092-4

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Conny Sjöqvist

Online version

Group Name: Molecular Ecology Lab
Subject: Marine Biology
University: Åbo Akademi University
Lab Website: connysjoqvist.com
AboCRIS Profile: research.abo.fi/en/persons/conny-sj%C3%B6qvist

Areas of Expertise

- Plankton ecology and evolution
- Experimental and molecular ecology
- Paleo- and resurrection biology

Research Projects

- Temperature adaptation in European diatom populations - AWARE
- Hotspots for biodiversity shifts in the Archipelago Sea - BIOSHIFT
- Modeling advanced primary production scenarios in coastal seas - MIMOSA
- Geochemical Dynamics of Seawater-Submarine Aquifer Interactions: Impacts on Coastal Sediments and Ecosystems
- Switching the functional roles of gene expression of feeding trait plasticity in a marine key species - GeneMac

Special Methodologies & Techniques

- Cell cultivation
- Microscopical analyses
- DNA and RNA sequencing
- Population genomics
- Transcriptomics
- Bioinformatics

Major Funding & International Networks

- Centre for Sustainable Ocean Science
- Swedish Cultural Foundation
- European Molecular Biology Laboratory (EMBL)
- Traversing European Coastlines (TREC)
- Estonian Research Council

Selected Publications

- Biodiversity of microorganisms in the Baltic Sea: the power of novel methods in the identification of marine microbes. FEMS Microbiology Reviews. 2024. DOI: 10.1093/femsre/fuae024
- Temperature optima of a natural diatom population increases as global warming proceeds. Nature Climate Change. 2024. DOI: 10.1038/s41558-024-01981-9
- Toward phytoplankton parasite detection using autoencoders. Machine Vision and Applications. 2023. DOI: 10.1007/s00138-023-01450-x
- Strain-specific transcriptional responses overshadow salinity effects in a marine diatom sampled along the Baltic Sea salinity cline. The ISME Journal. 2022. DOI: 10.1038/s41396-022-01230-x
- Ecologically coherent population structure of uncultivated bacterioplankton. The ISME Journal. 2021. DOI: 10.1038/s41396-021-00985-z

Last updated: 30 September 2026

MS

Martin Snickars

Online version

Group Name: Husö
Subject: Marine Biology
University: Åbo Akademi University
Lab Website: abo.fi/huso-biologiska-station
AboCRIS Profile: research.abo.fi/en/persons/martin-snickars

Areas of Expertise

- Coastal water ecology

Research Projects

- Marine Waterways -Together with local communities, the Åland Government, Åland Fisheries Center (Guttorp), SLU
- Currently involved in a large national biodiversity project focusing on the restoration of key coastal habitats.

Special Methodologies & Techniques

- Ordinary coastal monitoring methods and advanced echosounders and autonomous sonars

Major Funding & International Networks

- FIRI, EMBRC

Selected Publications

- Physical drivers of epi- and infauna communities related to dominating macrophytes in shallow bays in the Northern Baltic Sea. Estuarine, Coastal and Shelf Science. 2021. DOI: 10.1016/j.ecss.2021.107600
- Declines in predatory fish promote bloom-forming macroalgae. Ecological Applications. 2009. DOI: 10.1890/08-0964.1
- Impact of eutrophication and climate change on fish and zoobenthos in coastal waters of the Baltic Sea. Marine Biology. 2015. DOI: 10.1007/s00227-014-2579-3
- Essential coastal habitats for fish in the Baltic Sea. Estuarine, Coastal and Shelf Science. 2018. DOI: 10.1016/j.ecss.2018.02.014
- Coastal habitats and their importance for the diversity of benthic communities: A species- and trait-based approach. Estuarine, Coastal and Shelf Science. 2019. DOI: 10.1016/j.ecss.2019.106272

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Anna Törnroos-Remes

Online version

Group Name: Environmental and Marine Biology
Subject: Marine Biology
University: Åbo Akademi University
Lab Website: sites.abo.fi/centre-for-sustainable-ocean-science
AboCRIS Profile: research.abo.fi/sv/persons/anna-törnroos-remes

Areas of Expertise

- Marine community ecology
- Benthic ecology
- Marine biodiversity and ecosystem functioning
- Arctic benthic ecology, Interdisciplinary marine science

Research Projects

- Centre for Sustainable Ocean Science (www.abo.fi/sos) - an ÅAU internal Centre of Excellence (Center Lead)
- The Sea (www.abo.fi/sea) - an ÅAU research profile (PI, steering group)
- SEA-Quester (www.https://sea-quester.eu) - an EU Horizon Europe project (ÅAU PI, Task lead, WP co-task lead)
- PROTECT BALTIC (https://protectbaltic.eu)- an EU Horizon Europe project (ÅAU PI, Task lead)
- MARBEFES (https://marbefes.eu)- an EU Horizon Europe project (ÅAU PI, sub-task lead)
- Meyer Biodiversity project (ÅAU PI)

Special Methodologies & Techniques

- Trait-based approaches
- Ecological field experiments
- Reflexive learning
- Art-science approaches

Major Funding & International Networks

- Major funding sources: EU Horizon Europe, Research Council of Finland, Åbo Akademi University Foundation
- International networks: ICES WGIAB and WGBENT

Selected Publications

- Diverging benthic trait diversity and drivers across fjord to slope habitats of the high Arctic. Estuarine, Coastal and Shelf Science. 2025. DOI: 10.1016/j.ecss.2024.109086
- Introduction: towards holistic knowledge of marine environmental changes. Understanding Marine Changes. 2023. DOI: 10.4337/9781035311118.00006
- The anatomy of complex marine problems: a case study of decision-making on archipelagic aquaculture. Understanding Marine Changes. 2023. DOI: 10.4337/9781035311118.00013
- Biological traits approaches in benthic marine ecology: Dead ends and new paths. Ecology and Evolution. 2022. DOI: 10.1002/ece3.9001
- Four decades of functional community change reveals gradual trends and low interlinkage across trophic groups in a large marine ecosystem. Global Change Biology. 2019. DOI: 10.1111/gcb.14552

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Markus Öst

Online version

Group Name: Eider Research Team
Subject: Marine Biology
University: Åbo Akademi University
Lab Website: abo.fi/en/adaptations-to-challenging-environments-in-common-eiders
AboCRIS Profile: research.abo.fi/en/persons/markus-%C3%B6st

Areas of Expertise

- Population- and individual-level responses to increasing predation
- Fitness consequences of behavioural and physiological variation under predation and climate forcing
- Genetic connectivity, gene flow, and diversity during population decline
- Causes and consequences of breeding dispersal, choice of overwintering area, and migration
- Strategies, evolution of animal personality
- Life-history evolution
- Social evolution

Research Projects

- Ongoing project: Hormonal and genetic adaptations to a changing environment: a longitudinal population study of eiders (funding: Swedish Cultural Foundation in Finland)

Special Methodologies & Techniques

- Stress physiology: radioimmunoassay of corticosterone and other hormones (prolactin, thyroid hormones)
- Genetics: reduced-representation sequencing (RAD-seq)
- Movement: geolocator tracking
- Morphology and tagging: field measures and colour rings for individual identification

Major Funding & International Networks

- Academy of Finland, project 128039; 2009-2012.
- Swedish Cultural Foundation, 12 grants (2013-2024).
- Maj and Tor Nessling Foundation; 2009-2011.
- ARONIA 10-Year Jubilee Funds (2010).
- Academy of Finland project 104582; 2004-2007.
- Academy of Finland project 51895; 2001-2003.
- Delta Waterfowl Foundation (USA), 2004-2007.
- Oskar Öflunds stiftelse, 2006-2008.
- Ella och Georg Ehrnrooths stiftelse, 2001.

Selected Publications

- The effect of climate change on avian offspring production: A global meta-analysis. Proceedings of the National Academy of Sciences. 2023. DOI: 10.1073/pnas.2208389120
- Kin association during brood care in a facultatively social bird: active discrimination or by-product of partner choice and demography? Molecular Ecology. 2012. DOI: 10.1111/j.1365-294X.2012.05603.x
- Winter climate affects subsequent breeding success of common eiders. Global Change Biology. 2006. DOI: 10.1111/j.1365-2486.2006.01162.x
- Brood Size Matching: A Novel Perspective on Predator Dilution. The American Naturalist. 2013. DOI: 10.1086/668824
- Drivers of within- and among-individual variation in risk-taking behaviour during reproduction in a long-lived bird. Proceedings of the Royal Society B: Biological Sciences. 2022. DOI: 10.1098/rspb.2022.1338

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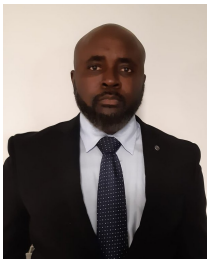


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Frederick Ahen

Online version

Group Name: Global Pharmacopolitics
Subject: Pharmacy
University: Åbo Akademi University
Lab Website: pharmscilab.fi
AboCRIS Profile: research.abo.fi/en/persons/frederick-ahen

Areas of Expertise

- Global health governance and pharmaceutical value creation
- Pharmaceutical Innovation Management
- Evidence-informed Pharmacopolitics and Pharmaceutical Policy
- Pharmastakeholder Engagement and Patient Outcomes
- Regulatory Affairs and Pharmacovigilance/anticounterfeits
- Sustainable Pharmacy and Health Equity
- Access to Medicines in Emerging Economies
- Pharmacognosy and markets for evidence-based herbal and alternative medicines

Research Projects

- The impact molecule
- Just sustainabilities in pharmaceutical ecosystems
- Adaptive pharmacopolitical Intelligence

Special Methodologies & Techniques

- Comparative institutional analysis
- Longitudinal Qualitative Methods
- Mixed Methods

Major Funding & International Networks

- Tor, Joe & Pentti Borg Memorial Fund
- Nordic POP (Patient-Oriented Products) University Hub https://nordicpop.ku.dk/
- Nordic Pharmaceutical Innovation and Translational (NordicPharmaTrain) network https://www.uu.se/en/department/medicinal-chemistry/faculty-of-pharmacy/nordicpharmatrain

Selected Publications

- Market violence through destructive entrepreneurship: Assessing institutional responses to the proliferation of counterfeit traditional and alternative medicines in Ghana. Heliyon. 2023. DOI: 10.1016/j.heliyon.2023.e13881
- Community-Level Health Interventions are Crucial in the Post-COVID-19 Era: Lessons from Africa's Proactive Public Health Policy Interventions. Humanistic Management Journal. 2022. DOI: 10.1007/s41463-022-00127-3
- From Ebola to COVID-19: what explains institutionalized manias and the ultimate preference for non-optimal solutions in global health governance? Critical Perspectives on International Business. 2021. DOI: 10.1108/cpoib-05-2020-0057
- Responsibilization and MNC–Stakeholder Engagement: Who Engages Whom in the Pharmaceutical Industry? Stakeholder Engagement: Clinical Research Cases. 2017. DOI: 10.1007/978-3-319-62785-4_5
- Critical perspectives on strategic CSR: what is sustainable value co-creation orientation? Critical Perspectives on International Business. 2015. DOI: 10.1108/cpoib-03-2012-0022

Last updated: 30 October 2026



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Kuldeep Bansal

Online version

Group Name: Polymeric drug delivery
Subject: Pharmacy
University: Åbo Akademi University
Lab Website: pharmscilab.fi
AboCRIS Profile: research.abo.fi/fi/persons/kuldeep-bansal

Areas of Expertise

- Functional Polymers
- Stimuli-Sensitive Polymers
- Targeted and Controlled Drug Delivery
- Polymeric Micelles
- Polymer-Drug Conjugates
- Microparticles
- Amorphous Solid Dispersions
- Polymeric Nanoemulsion

Research Projects

- Centre of Excellence in Materials-driven Solutions for Combatting Antimicrobial Resistance (MADNESS)
- Jasmine PRO: A versatile platform for drug delivery
- Non-Animal Platform for Nanoparticle-Based Delivery across the blood-brain barrier Interface with Vehicle Evolution

Special Methodologies & Techniques

- Polymer Synthesis and Characterization (NMR, FTIR, GPC, DSC),
- Nanoparticle Synthesis and Characterization (nanoprecipitation, microfluidics, DLS, HPLC, TEM),
- Cell Culture,
- Design of Stimuli-Sensitive Drug Delivery Platforms

Major Funding & International Networks

- Funding Sources: Stiftelsen för Åbo Akademi, Business Finland, Academy of Finland
- International Networks: Copenhagen University, NIPER (India), UiT The Arctic University of Norway, University College London

Selected Publications

- Reactive Oxygen Species-Regulated Conjugates Based on Poly(jasmine) Lactone for Simultaneous Delivery of Doxorubicin and Docetaxel. Pharmaceutics. 2024. DOI: 10.3390/pharmaceutics16091164
- A poly-δ-decalactone (PDL) based nanoemulgel for topical delivery of ketoconazole and eugenol against *Candida albicans*. Nanoscale Advances. 2024. DOI: 10.1039/D4NA00176A
- Utilizing the allyl-terminated copolymer methoxy(poly(ethylene glycol))-block-poly(jasmine lactone) in the development of amorphous solid dispersions: A comparative study of functionalized and non-functionalized polymer. International Journal of Pharmaceutics. 2024. DOI: 10.1016/j.ijpharm.2024.124175
- Functional block copolymer micelles based on poly (jasmine lactone) for improving the loading efficiency of weakly basic drugs. RSC Advances. 2022. DOI: 10.1039/D2RA03962A
- Synthesis and Evaluation of Novel Functional Polymers Derived from Renewable Jasmine Lactone for Stimuli-Responsive Drug Delivery. Advanced Functional Materials. 2021. DOI: 10.1002/adfm.202101998

Last updated: 30 September 2026

AF

Anne Filppula

Online version

Group Name: Translational Drug Research
Subject: Pharmacy
University: Åbo Akademi University
Lab Website: pharmscilab.fi/translationaladmeandpharmacodynamics
AboCRIS Profile: research.abo.fi/en/persons/anne-filppula

Areas of Expertise

- Pharmacokinetics, ADMET, drug interactions, pharmacogenetics, PBPK modelling

Research Projects

- Projects mainly related to drug interactions and pharmacokinetic modelling

Special Methodologies & Techniques

- PBPK modelling

Major Funding & International Networks

- Borgs fond, Liv & Hälsa

Selected Publications

- Candesartan Has No Clinically Meaningful Effect on the Plasma Concentrations of Cytochrome P450 2C8 Substrate Repaglinide in Humans. Drug Metabolism and Disposition. 2024. DOI: 10.1124/dmd.124.001798
- Hydroxychloroquine is Metabolized by Cytochrome P450 2D6, 3A4, and 2C8, and Inhibits Cytochrome P450 2D6, while its Metabolites also Inhibit Cytochrome P450 3A in vitro. Drug Metabolism and Disposition. 2023. DOI: 10.1124/dmd.122.001018
- Translational aspects of cytochrome P450-mediated drug–drug interactions: A case study with clopidogrel. Basic & Clinical Pharmacology & Toxicology. 2022. DOI: 10.1111/bcpt.13647
- Comparative Hepatic and Intestinal Metabolism and Pharmacodynamics of Statins. Drug Metabolism and Disposition. 2021. DOI: 10.1124/dmd.121.000406
- Improved predictions of time-dependent drug-drug interactions by determination of cytosolic drug concentrations. Scientific Reports. 2019. DOI: 10.1038/s41598-019-42051-x

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Zhang Hongbo

Online version

Group Name: Functional Materials for Medicine
Subject: Pharmacy
University: Åbo Akademi University
Lab Website: pharmscilab.fi/nanoprecisionmed
AboCRIS Profile: research.abo.fi/en/persons/hongbo-zhang

Areas of Expertise

- Nanomedicine
- Biomedical Engineering
- Drug Delivery
- Gene Therapy

Research Projects

- Functional Materials have played a more and more critical role in developing novel treatment methods for different diseases. Our group focuses on designing all kinds of functional materials, at different scales, to solve clinical and biological problems.

Special Methodologies & Techniques

- We have broad collaborations with hospitals, and most of our projects have clinical doctors involved, and initially, from a real-life clinical question. Our group is multidisciplinary, which involves an interplay between physics, materials science, synthetic organic chemistry, engineering, biology, and medicine. We aim that many of our designs could be clinically applicable in the future, and have a tremendous social impact in treating diseases.

Major Funding & International Networks

- I am coordinating many projects from the Research Council of Finland, Business Finland, and different foundations. I am also part of many consortium projects, including the InFlame flagship, NAP4DIVE (EU Horizon), Nordic POP, NordicPharmTrain, PII: Printed Intelligence Infrastructure, MADNESS (ÅAU Center of Excellence). As well as the collaboration project with companies, e.g., the joint project with Bayer Oy. I collaborate broadly with Harvard University, USA, and Chinese hospitals.

Selected Publications

- High-sensitivity, protein-independent detection of dsDNA sequences. Proceedings of the National Academy of Sciences. 2026. DOI: 10.1073/pnas.2515765123
- Disrupting Complement-Inflammation Positive Feedback Circuit via Oligonucleotide Hydrogel Microspheres for Reversing Joint Inflammation. Advanced Materials. 2026. DOI: 10.1002/adma.202518378
- Mitochondria-Targeted Nanomotor: H_2S-Driven Cascade Therapy for Hepatocellular Carcinoma. Advanced Materials. 2026. DOI: 10.1002/adma.202513757
- Artesunate Nanoplatform Targets the Serine–MAPK Axis in Cancer-Associated Fibroblasts to Reverse Photothermal Resistance in Triple-Negative Breast Cancer. Advanced Materials. 2025. DOI: 10.1002/adma.202502617
- Development of Aptamer-DNAzyme based metal-nucleic acid frameworks for gastric cancer therapy. Nature Communications. 2024. DOI: 10.1038/s41467-024-48149-9

Last updated: 30 September 2026

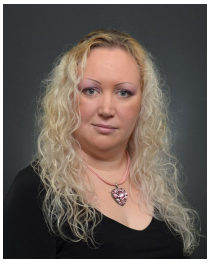


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Jessica Rosenholm

Group Name: BioNanoMaterials group
Subject: Pharmacy
University: Åbo Akademi University
Lab Website: pharmscilab.fi
AboCRIS Profile: research.abo.fi/en/persons/jessica-rosenholm

Online version

Areas of Expertise

- Development of novel drug delivery systems (DDS) based on new carrier materials e.g., naturally derived & synthetic polymers, inorganic materials, hydrogels, nanoparticulate systems
- Formulation of nanomedicines based on small and large bioactive molecules (small molecule drugs, peptides, proteins, nucleic acids)
- Rational DDS design through physico-chemical characterization linking material attributes to biobehavior, including cellular and bacterial assays

Research Projects

- https://research.abo.fi/en/persons/jessica-rosenholm/projects/

Special Methodologies & Techniques

- Nanotechnologies for drug delivery and imaging
- 2D and 3D printing technologies
- Physico-chemical characterization of nanomaterials

Major Funding & International Networks

- EU consortium "Non-Animal Platform for Nanoparticle-Based Delivery across the blood-brain barrier Interface with Vehicle Evolution" (NAP4DIVE) https://nap4dive.eu/
- Centre of Excellence in "Materials-driven Solutions for Combatting Antimicrobial Resistance" (MADNESS) https://sites.abo.fi/madness/
- Nordic POP (Patient-Oriented Products) University Hub https://nordicpop.ku.dk/
- Nordic Pharmaceutical Innovation and Translational (NordicPharmaTrain) network https://www.uu.se/en/department/medicinal-chemistry/faculty-of-pharmacy/nordicpharmatrain
- Finnish Platform for Pharmaceutical Research (FinPharma) https://www.helsinki.fi/en/networks/finpharma

Selected Publications

- Functional enzyme delivery via surface-modified mesoporous silica nanoparticles in 3D printed nanocomposite hydrogels. European Journal of Pharmaceutical Sciences. 2025. DOI: 10.1016/j.ejps.2025.107132
- Surface Modification of Mesoporous Silica Nanoparticles as a Means to Introduce Inherent Cancer-Targeting Ability in a 3D Tumor Microenvironment. Small Science. 2024. DOI: 10.1002/smsc.202400084
- Evolving Technologies and Strategies for Combating Antibacterial Resistance in the Advent of the Postantibiotic Era. Advanced Functional Materials. 2020. DOI: 10.1002/adfm.201908783
- Nanodiamonds for advanced optical bioimaging and beyond. Current Opinion in Colloid & Interface Science. 2019. DOI: 10.1016/j.cocis.2019.02.014
- Multimodality Imaging of Silica and Silicon Materials In Vivo. Advanced Materials. 2018. DOI: 10.1002/adma.201703651

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Outi Salo-Ahen

Online version

Group Name: Computer-aided drug design (CADD) group
Subject: Pharmacy
University: Åbo Akademi University
Lab Website: pharmscilab.fi/computer-aided-drug-design
AboCRIS Profile: research.abo.fi/sv/persons/outi-salo-ahen

Areas of Expertise

- Computer-aided drug design
- Molecular modeling
- Biomolecular simulations
- Computational chemistry/biology and computational pharmaceutics
- Structural bioinformatics
- Immunoinformatics

Research Projects

- Discovery and design of novel antivirulence agents / antimicrobial compounds / anticancer drugs
- Computational analysis of pharmaceutical materials (e.g., nanoparticles, polymers)

Special Methodologies & Techniques

- Comparative protein modeling
- Molecular docking
- Molecular dynamics simulations
- Molecular interaction analysis
- Materials science modeling

Major Funding & International Networks

- Tor, Joe & Pentti Memorial Fund
- Svenska Kulturfonden
- Research Council of Finland
- EU-OPENSCREEN

Selected Publications

- Microfluidics-enabled core/shell nanostructure assembly: Understanding encapsulation processes via particle characterization and molecular dynamics. Advances in Colloid and Interface Science. 2025. DOI: 10.1016/j.cis.2025.103400
- Insights into Molecular Interactions and Biological Effect of Natural Stilbenoids at the TRPA1 Ion Channel. ChemMedChem. 2025. DOI: 10.1002/cmdc.202400501
- Development of Aptamer-DNAzyme based metal-nucleic acid frameworks for gastric cancer therapy. Nature Communications. 2024. DOI: 10.1038/s41467-024-48149-9
- Isolation and functional analysis of phage-displayed antibody fragments targeting the staphylococcal superantigen-like proteins. MicrobiologyOpen. 2023. DOI: 10.1002/mbo3.1371
- The discovery of Zika virus NS2B-NS3 inhibitors with antiviral activity via an integrated virtual screening approach. European Journal of Pharmaceutical Sciences. 2022. DOI: 10.1016/j.ejps.2022.106220

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Tapani Viitala

Group Name: Pharmaceutical Biophysics
Subject: Pharmacy
University: Åbo Akademi University
AboCRIS Profile: research.abo.fi/fi/persons/tapani-viitala

Online version

Areas of Expertise

- Pharmaceutical nanotechnology
- Surface and colloid chemistry
- Physico-chemical characterization
- Drug delivery and targeting
- Biomolecular interactions
- Real-time label-free living cell sensing
- 3D printing

Research Projects

- NAP4DIVE: Non-Animal Platform for Nanoparticle-Based Delivery Across the Blood-Brain Barrier Interface with Vehicle Evolution - EU Horizon RIA
- Nordic Pharmaceutical Translation and Innovation - Nordforsk
- MADNESS: Centre of Excellence in Materials-driven solutions for combating antimicrobial resistance - Åbo Akademi Foundation
- Data integrated platforms for the design, production, and testing of therapeutics (project in Helsinki) - Business Finland

Special Methodologies & Techniques

- Multi-Parametric Surface Plasmon Resonance
- Impedance-based Quartz Crystal Microbalance
- LigandTracer
- Waveguide scattering microscopy

Major Funding & International Networks

- EU Horizon RIA
- NordForsk
- Åbo Akademi Foundation
- Business Finland

Selected Publications

- Semi-solid extruded tablets for personalized pediatric use: Development, Quality control and In-Vitro Assessment of Enteral Tube Administration. European Journal of Pharmaceutical Sciences. 2025. DOI: 10.1016/j.ejps.2025.107122
- Monitoring silica core@shell nanoparticle-bacterial film interactions using the multi-parametric surface plasmon resonance technique. Smart Medicine. 2023. DOI: 10.1002/SMMD.20230012
- *In Vitro* Characterization and Real-Time Label-Free Assessment of the Interaction of Chitosan-Coated Niosomes with Intestinal Cellular Monolayers. Langmuir. 2023. DOI: 10.1021/acs.langmuir.3c00728
- Protein A/G-based surface plasmon resonance biosensor for regenerable antibody-mediated capture and analysis of nanoparticles. Colloids and Surfaces A: Physicochemical and Engineering Aspects. 2022. DOI: 10.1016/j.colsurfa.2022.130015
- *In situ* analysis of liposome hard and soft protein corona structure and composition in a single label-free workflow. Nanoscale. 2020. DOI: 10.1039/C9NR08186K

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Xiaoju Wang

Online version

Group Name: Biomaterials and Drug Fabrication
Subject: Pharmacy
University: Åbo Akademi University
Lab Website: pharmscilab.fi/biomaterials-and-drug-fabrication
AboCRIS Profile: research.abo.fi/sv/persons/xiaoju-wang

Areas of Expertise

- My research focuses on harnessing the benign properties of bio-enabled plant biopolymers to design sustainable and functional biomaterials for better interfacing with living cells or microbes. To create effective bionanomaterials and soft matter biomaterials, my research applies core methodologies such as biopolymer chemical modifications, nanostructured biopolymer synthesis, and self-assembled biocolloidal networks from nano- to macro-scale.

Research Projects

- MADNESS CoE-Center of Excellence in Materials-driven solutions for combating antimicrobial resistance, Åbo Akademi University Foundation, 01.01.2024-31.12.2028 (Co-PI; Center director: Prof. Jessica Rosenholm)
- SusCellInk-Sustainable nanocellulose-based bioinks towards diverse material functionalities and therapeutic delivery of bioactive cues, Research Council of Finland (333158), 01.09.2020-31.08.2025, (PI)

Special Methodologies & Techniques

- Chemical synthesis of nanostructured polysaccharides and polyphenols
- Rheology
- Various additive manufacturing techniques
- Cell-laden 3D bioprinting for fabrication of in vitro models

Major Funding & International Networks

- Independent PI at CoE-MADNESS, Åbo Akademi University Foundation, 01.01.2024-31.12.2028
- Academy Research Fellow, Research Council of Finland (333158), 01.09.2020-31.08.2025

Selected Publications

- Photocurable cellulose nanofibers and their copolymers with polyacrylamide as microgels to support 3D cell cultivation. Nanoscale. 2025. DOI: 10.1039/D5NR00583C
- Bioprinting Macroporous Hydrogel with Aqueous Two-Phase Emulsion-Based Bioink: In Vitro Mineralization and Differentiation Empowered by Phosphorylated Cellulose Nanofibrils. Advanced Functional Materials. 2024. DOI: 10.1002/adfm.202400431
- Aqueous Two-Phase Emulsion Bioresin for Facile One-Step 3D Microgel-Based Bioprinting. Advanced Healthcare Materials. 2023. DOI: 10.1002/adhm.202203243
- Evolution of Self-Assembled Lignin Nanostructure into Dendritic Fiber in Aqueous Biphasic Photocurable Resin for DLP-Printing. Advanced Functional Materials. 2024. DOI: 10.1002/adfm.202315679
- Digital light processing (DLP) 3D-fabricated antimicrobial hydrogel with a sustainable resin of methacrylated woody polysaccharides and hybrid silver-lignin nanospheres. Green Chemistry. 2022. DOI: 10.1039/D1GC03841A

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